

Find angles in triangles and quadrilaterals

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A triangle has angles 47 degrees and 68 degrees. Find the third angle. Copy or complete the first step.

2. A circle has radius 4.5 cm. What is its diameter?

3. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A triangle has angles 47 degrees and 68 degrees. Find the third angle. Copy or complete the first step.
Answer 65 degrees.
2. A circle has radius 4.5 cm. What is its diameter?
Answer 9 cm.
3. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer 75 degrees.
4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.
Answer Use labelled values and known geometric facts, then check the total angle relationship.

Find angles in triangles and quadrilaterals

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A triangle has angles 47 degrees and 68 degrees. Find the third angle.

6. Check this result using an inverse, estimate or known fact: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

2. A circle has radius 4.5 cm. What is its diameter?

7. Prove or explain your reasoning: Angles around a point are 85 degrees, 140 degrees and x . Find x .

3. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.

4. Solve this independently and show every stage: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A circle has radius 4.5 cm. What is its diameter?

10. Write and solve a similar question to: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

Teacher answers and checking prompts

1. A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees.
2. A circle has radius 4.5 cm. What is its diameter?
Answer 9 cm.
3. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer 75 degrees.
4. Solve this independently and show every stage: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees.
5. Use a second method or representation for: A circle has radius 4.5 cm. What is its diameter?
Answer Expected result: 9 cm. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer Expected result: 75 degrees. A valid check must be shown.
7. Prove or explain your reasoning: Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer 135 degrees. The explanation must show why the method works.
8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.
Answer Use labelled values and known geometric facts, then check the total angle relationship.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Find angles in triangles and quadrilaterals

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Angles around a point are 85 degrees, 140 degrees and x . Find x .

6. Create a new example that tests the same idea as: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

2. Prove the answer to this question, rather than only calculating it: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A circle has radius 4.5 cm. What is its diameter?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 75 degrees.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer 135 degrees. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer Expected result: 65 degrees. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A circle has radius 4.5 cm. What is its diameter?
Answer Expected result: 9 cm. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 75 degrees.
Answer One valid reconstruction is: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.
Answer Use labelled values and known geometric facts, then check the total angle relationship.
6. Create a new example that tests the same idea as: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.